

Machine Learning Analysis Report

Occupation Classification and Gender Bias in Professional Biographies

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May 1, 2026

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1 Overview

This project trains a 28-class occupation classifier on professional biography text using the BiasBios dataset (De-Arteaga et al., 2019), as reconstructed and released by Ravfogel et al. (2020) and hosted on Hugging Face (Curien, 2020). Logistic regression was chosen because its weight matrix is directly readable as a mechanism: inspecting it requires no separate probe or approximation. Weight inspection revealed gender signal concentrated in a small number of features, enabling a targeted intervention: removing nine pronoun tokens reduced mean gender correlation by 40% at no cost to macro F1. This interpretability is a consequence of model capacity rather than deliberate design, and its limits as complexity grows are the central question of the broader project arc.

2 Dataset Description

De-Arteaga et al. (2019) compiled a dataset of professional biographies, along with their occupations and genders, from a web crawl. Although the crawling code was released, the dataset itself was not. Ravfogel et al. (2020) released a reconstructed version for their own experiments, which is the version used here, hosted on Hugging Face by Laboratoire Hubert Curien (Curien, 2020). The reconstructed version contains 396,189 biographies across 28 occupation classes, slightly smaller than the original due to 5,557 biographies no longer available on OpenCrawl at the time of reconstruction.

The dataset contains four columns: **hard_text** (biography text), **profession** (integer-encoded occupation class), **gender** (binary integer: 0 for male, 1 for female), and **split** (train, dev, or test). The predefined splits from Ravfogel et al. (2020) are preserved exactly: 257,478 biographies in train, 39,642 in dev, and 99,069 in test. The dev set is used for hyperparameter tuning and the test set is held out until final evaluation. The **gender** column is not used as a model input; it is retained exclusively for the ethics analysis. Although integer encoding aids training, it hampers interpretation: two columns were added during preprocessing, **profession_text** and **gender_text**, containing the corresponding textual labels for each integer code.

3 Modeling Approach

The modeling pipeline begins with feature engineering. Biographies are converted to numerical vectors using TF-IDF, the standard bag-of-words vectorisation method in classical NLP (Schütze et al., 2008). Compared to raw count vectors, TF-IDF’s IDF weighting suppresses terms common across documents while sublinear TF normalisation reduces the dominance of high-frequency terms, making it strictly preferable for text classification. The vocabulary is restricted to terms appearing in at least 0.1% and at most 99% of documents, yielding 4,402 active features after filtering. Stopword removal is deliberately skipped to preserve explicit gender indicators such as *he* and *she*, which are central to the ethics analysis. The vectoriser is fitted on the training set only to prevent data leakage.

Logistic regression was chosen as the classifier. Beyond its computational efficiency, its coefficient matrix is directly readable as a mechanism: each weight encodes the contribution of a single feature to a single class prediction, requiring no post-hoc approximation to interpret. This makes logistic regression the degenerate case of a linear probe (Alain and Bengio, 2016), where the probe and the model are identical. However, this interpretability is a consequence of the model’s limited capacity: the same linearity that makes the weights readable also bounds expressiveness, and performance on most tasks falls short of modern deep learning.

The only hyperparameter requiring tuning is C, the inverse regularisation strength, which directly controls the bias-variance tradeoff and is sensitive to feature space scale (Hastie, 2009). A grid

search over candidate values was conducted, scoring on macro-averaged F1 to weight all 28 classes equally regardless of size. To avoid overfitting to the test set, tuning used the predefined dev split from Ravfogel et al. (2020) via `PredefinedSplit`. The search identified $C = 1.0$ as optimal. The model was trained using the L-BFGS solver for up to 1,000 iterations with balanced class weighting, which adjusts loss contributions inversely proportional to class frequency. Given the strong overrepresentation of professors (29,520 test examples vs. 351 for rapper), balanced weighting prevents the model from defaulting to majority-class predictions at the expense of rarer occupations.

A `DummyClassifier` predicting the most frequent class at all times serves as the baseline, yielding a macro F1 of 0.02 and an accuracy of 30%. This establishes the floor against which the logistic regression model is evaluated.

4 Results

The logistic regression model achieves 76% accuracy and a macro-averaged F1 of 0.68 on the test set, compared to the majority-class baseline accuracy of 30% and macro F1 of 0.02. The improvement of 0.66 macro F1 points demonstrates that the model has learned meaningful discriminative features across professions rather than exploiting class imbalance.

Macro-averaged F1 is the primary evaluation metric, weighting all 28 classes equally regardless of size. Accuracy alone would be misleading given the strong overrepresentation of professors (29,520 test examples). Table 1 summarises the overall results and Table 2 highlights the best and worst performing classes.

Model	Accuracy	Macro F1
DummyClassifier (most frequent)	0.30	0.02
Logistic Regression	0.76	0.68

Table 1: Classification performance on the test set (n = 99,069).

Profession	Precision	Recall	F1	Support
dentist	0.93	0.92	0.93	3,647
attorney	0.89	0.83	0.86	8,143
photographer	0.89	0.80	0.85	6,068
nurse	0.81	0.83	0.82	4,738
pastor	0.31	0.74	0.44	632
interior_designer	0.37	0.73	0.49	366
paralegal	0.45	0.75	0.56	442

Table 2: Best and worst performing classes on the test set.

The best performing classes all have distinctive occupational vocabulary. The weakest class is pastor (F1 = 0.44), a surprising result given that clerical biographies might be expected to contain relatively distinctive language. Interior designer (0.49) and paralegal (0.56) likely suffer from vocabulary overlap with adjacent professions. The consistent recall scores across classes confirm that balanced class weighting is effective at preventing majority-class dominance.

5 Ethics, Interpretability and Bias

In mechanistic interpretability, linear probes are trained on top of a model’s internal representations to test what information is encoded at a given layer (Alain and Bengio, 2016). The probe is a separate, lightweight classifier layered on top of an opaque model to recover what the

model itself cannot explain. Logistic regression does not require this indirection. The weight matrix directly maps TF-IDF features to class predictions, meaning the model’s decision logic is fully exposed without any additional machinery. This is not a deliberate design choice for interpretability; it is a consequence of the model’s limited capacity. The same linearity that constrains expressiveness makes the internal logic directly readable: the weights are the probe.

The model has learned 4,402 weights for each of the 28 professions, expressed as the softmax function:

$$P(y = k \mid \mathbf{x}) = \frac{e^{\mathbf{w}_k \cdot \mathbf{x} + b_k}}{\sum_{j=1}^K e^{\mathbf{w}_j \cdot \mathbf{x} + b_j}}$$

This is the same output layer used in modern neural classifiers. A feature drives a prediction when two conditions hold: its weight is large, and its presence in the input is strong. The relevant quantity is therefore the logit contribution, the product of weight and input signal strength, rather than the weight alone.

The top 25 features by weight are extracted per occupation class and their logit contributions are correlated with the binary gender label using the point-biserial correlation coefficient (Lev, 1949), the appropriate statistic when one variable is continuous and the other is binary, mathematically equivalent to Pearson’s r in this case. The sign is directly interpretable: positive values indicate female-correlated features, negative values indicate male-correlated features. De-Arteaga et al. (2019) identified gender bias in this dataset; the weight inspection here confirms and localises it.

As shown in Figure 1, the pattern is strikingly symmetric: *she* dominates female-correlated professions including dietitian, nurse, yoga_teacher, model, and teacher, while *he* dominates male-correlated professions including chiropractor, surgeon, software_engineer, comedian, and pastor, both at absolute correlations of approximately 0.75. In the middle tier, occupation-specific terms such as *nursing*, *dr*, and *software* dominate, suggesting the model relies on occupational vocabulary rather than gendered pronouns for these classes. Figure 2 confirms that gender signal is concentrated in pronouns rather than distributed across the weight vector: correlations drop sharply at the second-ranked feature, where occupation-specific terms dominate instead.

This localisation enables a targeted intervention. A pronoun stop list of nine tokens (*she*, *her*, *hers*, *he*, *his*, *him*, *ms*, *mr*, *mrs*) is added to the TF-IDF vectoriser, with all other hyperparameters held constant. Non-binary pronouns (*they*, *them*, *their*) are deliberately excluded as they are common in generic English usage. The two models are compared on macro F1 and on mean absolute point-biserial correlation, defined as:

$$\bar{r} = \frac{1}{C} \sum_{c=1}^C \frac{1}{K} \sum_{k=1}^K |r_{c,k}|$$

where $C = 28$ occupation classes, $K = 25$ top-weighted features per class, and $r_{c,k}$ is the point-biserial correlation between the logit contribution of feature k in class c and the binary gender label.

The intervened model achieves a macro F1 of 0.68, identical to the baseline, confirming that occupational predictive signal is carried by content vocabulary rather than gendered pronouns. Mean $\bar{r}$ falls from 0.035 to 0.021, a reduction of 40%, as shown in Figure 3. Residual gendered signal remains: *wife* and *husband* appear as top features in the intervened model, indicating that gendered language extends beyond explicit pronouns. A more complete debiasing would require suppressing gender information directly in the representation space, an approach that becomes necessary as model complexity grows and the weight matrix is no longer directly readable.

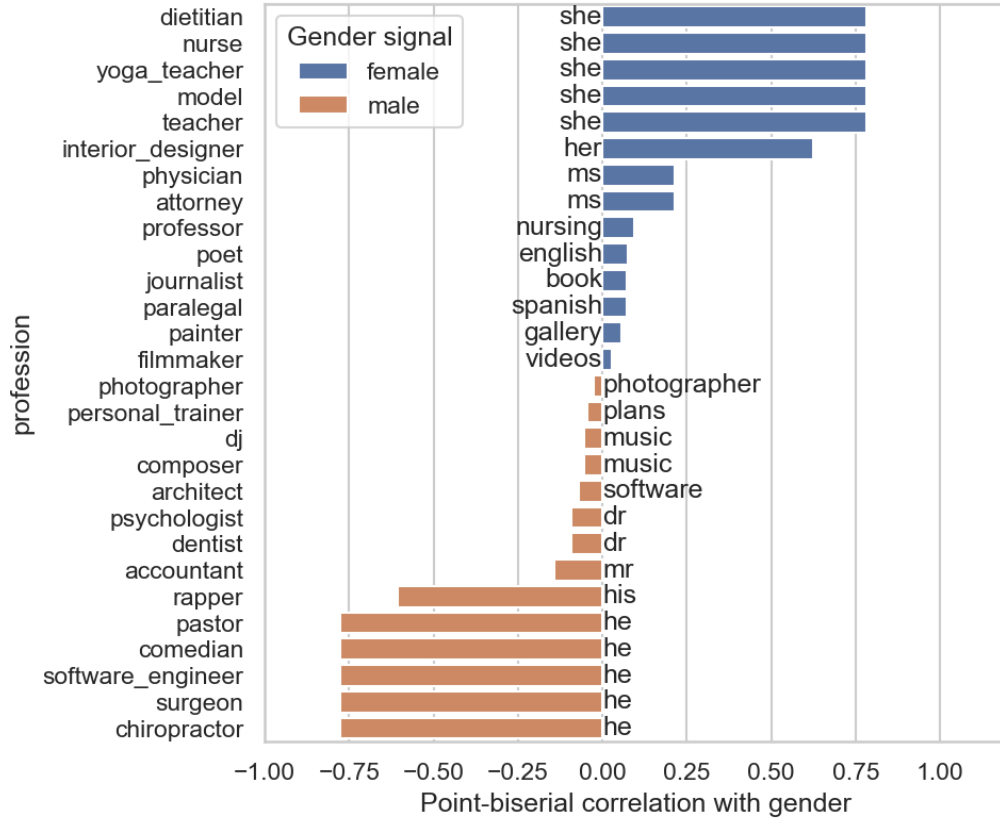


Figure 1: Top-ranked gender-correlated feature per profession in the baseline model. Bar width indicates point-biserial correlation strength; colour indicates direction (female/male). Feature name is displayed next to each bar.

Finally, it is worth acknowledging the researcher’s own positionality. The framing of this analysis assumes that gender should not be determinative of professional classification. This is less an empirical claim than a moral and political conviction, and one that may not hold uniformly across cultural contexts. What is interpreted as bias here reflects a value judgment shared widely within the AI research community (De-Arteaga et al., 2019), but not universally. In the interest of the same transparency this project advocates for in models, it is worth making that assumption explicit: the reader should know the normative position from which the analysis proceeds.

6 Interpretation for a Non-Technical Audience

As the prevalence and capabilities of AI systems increase, our ability to understand what they are thinking and why they make their decisions is fading. Experts call the field that fights against this loss of interpretability Explainable AI. This analysis shows why understanding AI is of significant importance to its ethical employment.

The Bias in Bios dataset contains biographies compiled from the Internet, and its goal is to classify the profession of the person behind each biography. However, the dataset comes with a twist: it has a known gender bias and metadata indicating the biography’s gender. The question is whether a machine learning model trained on this biased dataset will learn a gender bias towards certain professions. Should identifying gender matter when classifying someone as a model, a teacher, or a software engineer, or does this very act propagate stereotypes?

For the analysis, a classical machine learning model called logistic regression was trained. The

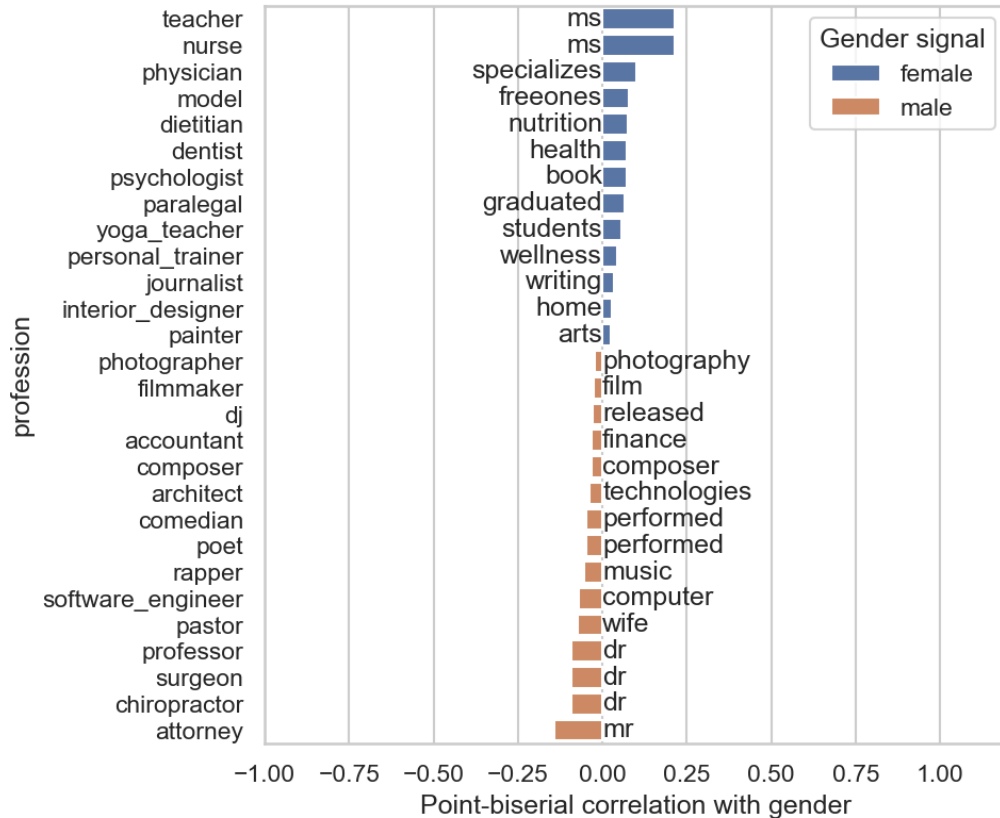


Figure 2: Second-ranked gender-correlated feature per profession in the baseline model. Correlations drop sharply relative to Figure 1, confirming that gender signal is concentrated rather than distributed.

reason is that it is easier to deduce the decision processes behind classical machine learning models than those of complex modern AI systems. What we found is a clear association of the word *she* with nurses, teachers, models, dietitians, and yoga instructors. On the other hand, when *he* is frequent in a biography, the model was more likely to assign professions such as chiropractor, surgeon, software engineer, comedian, or pastor. The bias present in the biographies was learned by the model.

The gendering of professions is a clear concern for perpetuating stereotypes that might reinforce patterns of a gendered society. An intervention was therefore conducted: pronouns were eliminated from the vocabulary fed into the model. This experiment showed that a model trained without pronouns was substantially less gender-biased than the original. What was surprising, however, was that its performance did not degrade. It was just as good at predicting professions as the more biased model.

What this shows is that there is a choice between training a gender-biased model and a far less gender-biased one, without sacrificing usability. Through the model’s interpretability, it is sometimes possible to identify and reduce bias precisely. If we can find ways to apply this kind of analysis not just to simple classical models but also to modern AI systems, we may be able to make them more equitable. This idea will be explored further in subsequent projects.

7 Limitations and Potential Bias

The first limitation concerns the TF-IDF representation. As a bag-of-words method, it counts terms with frequency weighting but discards semantics, word order, and co-reference (Schütze

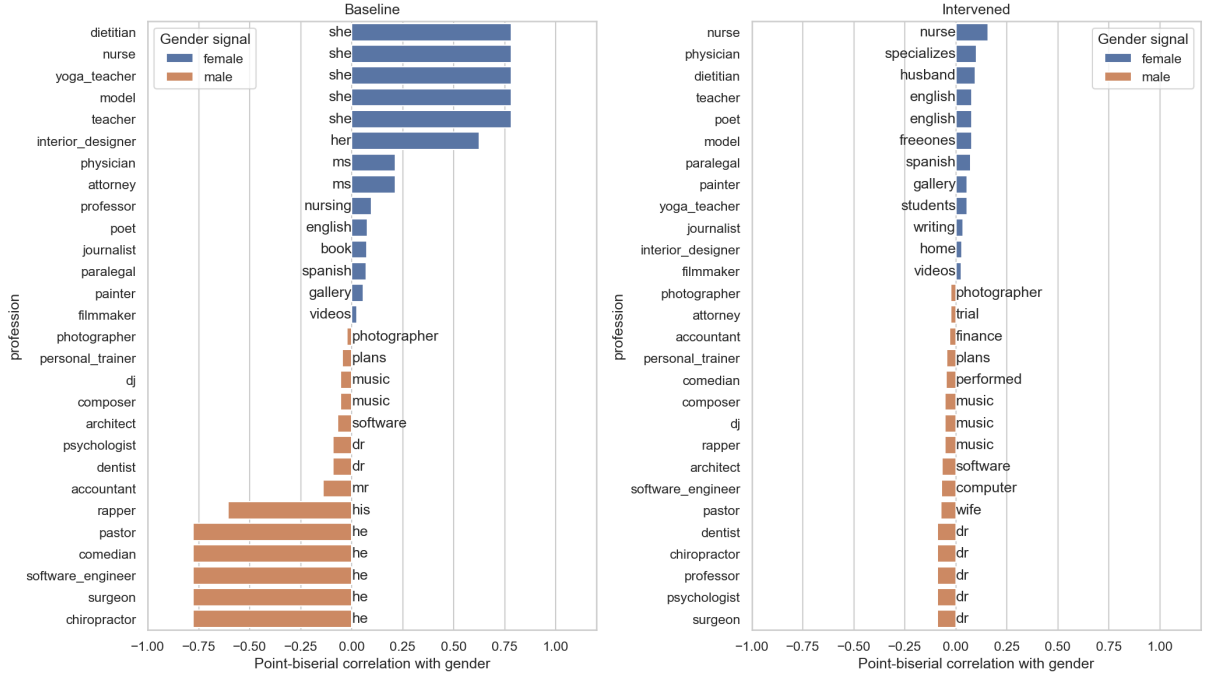


Figure 3: Baseline (left) vs. intervened model (right): top-ranked gender-correlated feature per profession. After removing nine pronoun tokens, mean $\bar{r}$ falls from 0.035 to 0.021 (40% reduction) with no change in macro F1.

et al., 2008). The pronoun *she*, for example, may refer to the subject of the biography or to a third party mentioned in it, so the model cannot distinguish between these cases. Semantically richer representations such as learned word embeddings or transformer-based encoders would address this, but at the cost of the direct weight interpretability that this analysis depends on.

Second, the dataset originates from a web crawl and is known to over-represent academic biographies, with professors comprising 30% of the test set (De-Arteaga et al., 2019). This sampling bias means the dataset does not represent the true distribution of professions in the population. A more representative dataset would require access to private documents such as CVs, which cannot be used in a reproducible academic setting. Subsampling or oversampling to better approximate the true distribution was out of scope for this project, given that the primary goal is to demonstrate how interpretability can expose and reduce bias rather than to maximise classification performance.

Third, the pronoun intervention is only partially effective. Tokens such as *wife* and *husband* survive the stop list and carry residual gender signal, confirming that gendered language is not limited to explicit pronouns. A more complete debiasing would require either a broader intervention or a method that suppresses gender information directly in the representation space.

Most fundamentally, the interpretability demonstrated here is a consequence of model capacity rather than a designed property (Alain and Bengio, 2016). Logistic regression exposes its decision logic through the weight matrix because it has no intermediate representations. As model capacity grows from classical classifiers to recurrent networks to transformers, this direct readability breaks down: gender signal can be reconstructed from subtler co-occurring vocabulary even without explicit pronouns, and it becomes encoded in intermediate representations that are not directly readable. Recovering interpretability at higher capacity requires dedicated tools such as linear probes and sparse autoencoders. This is the central question the subsequent projects in this capstone arc address.

8 Future Work

This project demonstrates that interpretability is tractable at the level of logistic regression precisely because the model has no intermediate representations. The weight matrix is the mechanism, and reading it requires no additional tools. As model capacity grows, this property disappears.

Subsequent projects in this capstone arc investigate what happens at higher complexity. Recurrent networks and transformers encode information in intermediate representations that are not directly readable, and gender signal can be reconstructed from subtle co-occurring vocabulary even after explicit pronouns are removed. Recovering interpretability in these settings requires dedicated methods: linear probes (Alain and Bengio, 2016) to test what information is encoded at a given layer, and sparse autoencoders to recover monosemantic features from superposed representations. The question of whether bias can still be identified and reduced under these conditions, and at what cost, is the thread that connects this project to the rest of the arc.

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